

- Write a program to calculate the n -th Fibonacci number. The recursive formula for the Fibonacci sequence is:

$$F_n := F(n) := \begin{cases} 0, & \text{when } n = 0; \\ 1, & \text{when } n = 1; \\ F(n-1) + F(n-2) & \text{when } n > 1. \end{cases}$$

- The least common multiple (LCM) of two integers a and b is the smallest positive integer that is divisible by both a and b . LCM has a close relationship with GCD through the following formula:
 $lcm(a, b) = (|a * b|) / gcd(a, b)$. Write a function to compute the LCM of any two numbers.
- Write the function to calculate the sum:

$$S = \frac{1}{2} + \frac{1}{2+4} + \dots + \frac{1}{2+4+\dots+(2*n)}$$

- Write a function to calculate the value of the following expression:

$$S = \underbrace{\sqrt{x + \sqrt{x + \sqrt{x + \dots + \sqrt{x + \sqrt{x}}}}}}_n$$